L2A: Substitution Model, Recursion

CS1101S: Programming Methodology

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Before we start...

Where are we in CS1101S?

What is CS1101S? SICPy, Source Academy, Studios, Missions, Ed, Paths

Getting into gear

- Path P2A: out today, to complete by tomorrow
- Check-In I2A: due during the lecture
- Reflection R2: sheet out on Canvas, please prepare before your reflection session tomorrow
- Mission M2A: out today after lecture
- Quest Q2A: out today evening
- Mission M2B: out tomorrow
- Quest Q2B: out on Friday
- You cannot attend a different Studio session if you miss your own

Earning eXperience Points

- Missions (basic homework): your primary source of XP
- Quests (optional homework): supplement your XP
- Early Mission/Quest submission bonus: 75 XP within 3 days;
  bonus decays after that
- To have your submission graded, you must **finalize** it in Source Academy, saving alone is not enough
- Finalize before the deadline to have it graded; Missions will be **auto-finalized** at the deadline
- Avengers can "unsubmit" your submission, within reasonable limits
- If you submit less than 2 days after release of Mission/Quest, Avengers will give feedback less than 4 days after release.
- Path: XP for submission
- Early path submission: 25 XP within 1 day; bonus decays after that
- No "unsubmit" for paths

Need help?

- Ed forums: `https://edstem.org/us/courses/101145/discussion`
- PM your Avenger
- **Forum Hours**. Every Tuesday 6:30pm-8:30pm, starting from Week 3. Venue is COM1 basement.
- **Reading Assessment 1**. Practice papers will be released next week!

Substitution model, 1.1.3 and 1.1.5

Recursion and iteration, 1.2.1

Substitution model, 1.1.3 and 1.1.5

- Evaluating combinations, 1.1.3
- Evaluating function application, 1.1.5
- Applicative vs normal order reduction, 1.1.5

Recursion and iteration, 1.2.1

Review: Evaluating arithmetic expressions, see 1.1.3

```
1 + 2 * 3 + 4
```

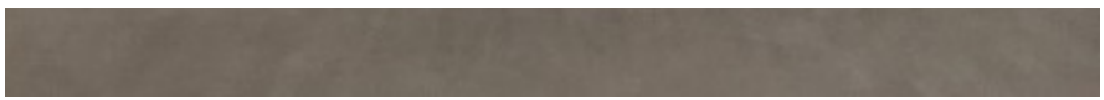
Replace "eligible" subexpression with result

```
(1 + (2 * 3)) + 4
(1 + 6) + 4
7 + 4
11
```

Example for substitution model 1.1.5

Problem

You need to find the cost of a circular handrail, knowing that the cost of the rail per meter is 199.95 dollars, and the radius of the circle is 2.1 meters.



Evaluation of function application (1.1.5)

```
cost_per_meter = 199.95
def circumference(radius):
    return 2 * math_pi * radius

def cost_of_circular_handrail(r):
    return cost_per_meter * circumference(r)

print(cost_of_circular_handrail(2.1))
```

```
-> 199.95 * circumference(2.1)
-> 199.95 * ((2 * 3.141592) * 2.1)
-> 199.95 * (6.283185 * 2.1)
-> 199.95 * 13.19469
-> 2638.278
```

Applicative order reduction, 1.1.5

```
def square(x):
    return x * x

def sum_of_squares(x, y):
    return square(x) + square(y)

def f(a):
    return sum_of_squares(a + 1, a * 2)

print(f(5))
```

```
->  sum_of_squares(5 + 1, 5 * 2)
->  sum_of_squares(6, 10)
->  square(6) + square(10)
->  (6 * 6) + (10 * 10)
...
->  136
```

Normal order reduction, 1.1.5

```
def square(x):
    return x * x

def sum_of_squares(x, y):
    return square(x) + square(y)

def f(a):
    return sum_of_squares(a + 1, a * 2)

print(f(5))
```

```
->  sum_of_squares(5 + 1, 5 * 2)
->  square(5 + 1) + square(5 * 2)
->  ((5 + 1) * (5 + 1)) +
    ((5 * 2) * (5 * 2))
...
->  136
```

No link? Python itself evaluates in applicative order, not the normal order shown in this trace.

Substitution model for applicative-order languages

Primitive expressions : take the value

Operator combinations : evaluate operands, apply operator

Declaration assignment : evaluate the value expression and replace the name by value

Function application : evaluate component expressions

- if function is primitive: apply the primitive function
- if function is compound: substitute argument values for parameters in body of declaration

Substitution model for runes

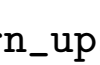
```
def turn_upside_down(rune):
    return quarter_turn_right(quarter_turn_right(rune))

def quarter_turn_left(rune):
    return quarter_turn_right(turn_upside_down(rune))

show(quarter_turn_left(heart))
```

Substitution model for runes

`quarter_turn_left(`  `)`

`quarter_turn_r(turn_upside_down(`  `))`

`quarter_turn_r(quarter_turn_r(quarter_turn_r(`  `)))`

`quarter_turn_r(quarter_turn_r(`  `))`

`quarter_turn_r(`  `)`

Recursion and iteration, 1.2.1

- `stackn` and `repeat_apply`
- Simple factorial and its recursive process, 1.2.1

A new primitive combination: `stack_frac`

`stack_frac(r, heart, sail)`

splits available bounded box such that heart occupies top fraction `r` of box and sail occupies remaining `1 - r` of box

Examples

```
stack_frac(87 / 100, heart, sail)
```

splits available bounded box such that heart occupies the top 87% of box and sail occupies remaining 13% of box

Trisection of the heart

```
show(stack_frac(1 / 3, heart,
    stack_frac(1 / 2, heart, heart)))
```

Can we define `stackn` in Python?

Trisection of the heart

```
show(stack_frac(1 / 3, heart,
    stack_frac(1 / 2, heart, heart)))
```

Quadrisection of the heart

```
show(stack_frac(1 / 4, heart,
    stack_frac(1 / 3, heart,
        stack_frac(1 / 2, heart, heart))))
```

Can we generalise this idea?

A Recursive Function, first try

```
def stackn(n, rune):
    return stack_frac(1 / n, rune,
        stackn(n - 1, rune))
```

Programs precisely describe a computational process.

We need to specify each step carefully and unambiguously.

The correct version

```
def stackn(n, rune):
    return rune if n == 1 else stack_frac(1 / n, rune,
        stackn(n - 1, rune))
```

Observation

Solution for n computed using solution $n - 1$,

solution for $n - 1$ is computed using solution $n - 2$, ...

until we reach trivial case.

A Recipe

```
def stackn(n, rune):
    return rune if n == 1 else stack_frac(1 / n, rune,
        stackn(n - 1, rune))
```

Recipe for recursion

- Figure out trivial *base case*
- Assume you know how to solve problem for $n-1$.
  How can we solve problem for n?

We call this **Wishful Thinking**. Make a wish and it will come true.

A Recipe

```
def stackn(n, rune):
    return rune if n == 1 else stack_frac(1 / n, rune,
        stackn(n - 1, rune))
```

Recipe for recursion

- Figure out trivial *base case*
- I **wish** that I already have the solution for a problem $n - 1$.
  Given that, how can I solve problem for n?

We call this **Wishful Thinking**. Make a wish and it will come true.

It's circles all the way down...

Let's circle back to the Source Academy!

Look for Check-In I2A in the Source Academy!

Can we define `repeat_apply` in Python?

Remember pre-defined `repeat_apply` in module `repeat`

```
show(repeat_apply(make_cross, 3, sail))
# should lead to
show(make_cross(make_cross(make_cross(sail))))
```

Another example

```
def square(x):
    return x * x

repeat_apply(square, 3, 2)
# should lead to
square(square(square(2)))
```

`repeat_apply`, our first version

```
def repeat_apply(pat, n, init):
    return init if n == 0 else pat(repeat_apply(pat, n - 1,
        init))
```

```
print(repeat_apply(square, 3, 2))
```

Recursive process

The applications of `pat` *accumulate* as result of recursive calls.
They are *deferred* operations.

`repeat_apply`, second version

```
def repeat_apply(pat, n, rune):
    return rune if n == 0 else repeat_apply(pat, n - 1,
        pat(rune))
```

```
print(repeat_apply(square, 3, 2))
```

Iterative process

With applicative order reduction, `pat` function is applied *before* the recursive call.

There are no deferred operations.

Another example: Factorial 1.2.1

Factorial

$n\,! = n(n-1)(n-2)\cdots 1$

Grouping

$n\,! = n((n-1)(n-2)\cdots 1)$

Replacement

$n\,! = n(n-1)\,!$

Remember the base case

$n\,! = 1 \qquad \text{if } n = 0$

$n\,! = n\,(n-1)\,! \qquad \text{if } n > 0$

Translation into Python

Remember the base case

$$
\begin{aligned}
n\,! &= 1 && \text{if } n = 0 \\
n\,! &= n(n-1)\,! && \text{if } n > 0
\end{aligned}
$$

Factorial in Python

```
def factorial(n):
    return 1 if n == 0 else n * factorial(n - 1)
```

Example execution using Substitution Model

```
def factorial(n):
    return 1 if n == 0 else n * factorial(n - 1)
```

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * (1 * factorial(0))))
4 * (3 * (2 * (1 * 1)))
4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
24
```

Notice the deferred operations.

Recursive process.

A Closer look at performance

Dimensions of performance

- Time:
  how long does the program run
- Space:
  how much memory do we need to run the program

Time for calculating $n!$

Executed Operations

Total number of executed operations

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * (1 * factorial(0))))
4 * (3 * (2 * (1 * 1)))
4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
24
```

Space for calculating $n!$

Deferred operations:

Number of "things to remember"

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * (1 * factorial(0))))
4 * (3 * (2 * (1 * 1)))
4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
24
```

Can we write an iterative factorial?

```
def factorial(n):
    return fact_iter(1, 0, n)

def fact_iter(product, counter, n):
    return product if counter == n else fact_iter(product *
    (counter + 1),
        counter + 1, n)
```

Summary

- Substitution allows us to trace the evaluation of expressions
  - Applicative vs normal order reduction
- Function applications are replaced by the return expression
- Examples for recursive solutions: `stackn`, `repeat_apply`, `factorial`
- Recursive and iterative processes
- Resources for computational processes:
  time and space
  (more on this on Friday)